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to chlorophyll 'a'. Chlorophyll molecules have a tadpole like structure. It consists of Mg-Porphyrin head (Hydrophilic Head) and (Lipophilic tail) Phytol tail. The Porphyrin head consists of four pyrrol rings linked together by C-H bridges. Each pyrrole ring comprises of four carbons and one nitrogen atom. Porphyrin ring has several side groups which alter the properties of the pigment. Different side groups are indicative of various types of chlorophyll. The Phytol tail made up of 20 carbon alcohol is attached to carbon 7 of the Pyrrole ring IV. It has a long propionic acid ester bond. Long lipophilic tail helps in anchoring chlorophyll to the lamellae (Figure 13.2).

i. Biosynthesis of Chlorophyll

Chlorophyll is synthesized from intermediates of respiration and photosynthesis. Succinic acid an intermediate of Krebs cycle is activated by the addition of coenzyme A and it reacts with a simple amino acid glycine and the reaction goes on to produce chlorophyll 'a'. Bio synthesis of chlorophyll a requires Mg, Fe, Cu, Zn, Mn, K and nitrogen. The absence of any one of these minerals leads to chlorosis (Recall what you have studied in '*Mineral Nutrition*').

ii. Comparison of Chlorophyll – a with other pigments

1. Chlorophyll 'b' differs from Chlorophyll 'a' in having CHO (aldehyde) group instead of CH_3 (Methyl) group at the 3rd C atom in II Pyrrol ring (Figure 13.2).
2. Chlorophyll 'c' differs from Chlorophyll 'a' by lacking phytol tail.

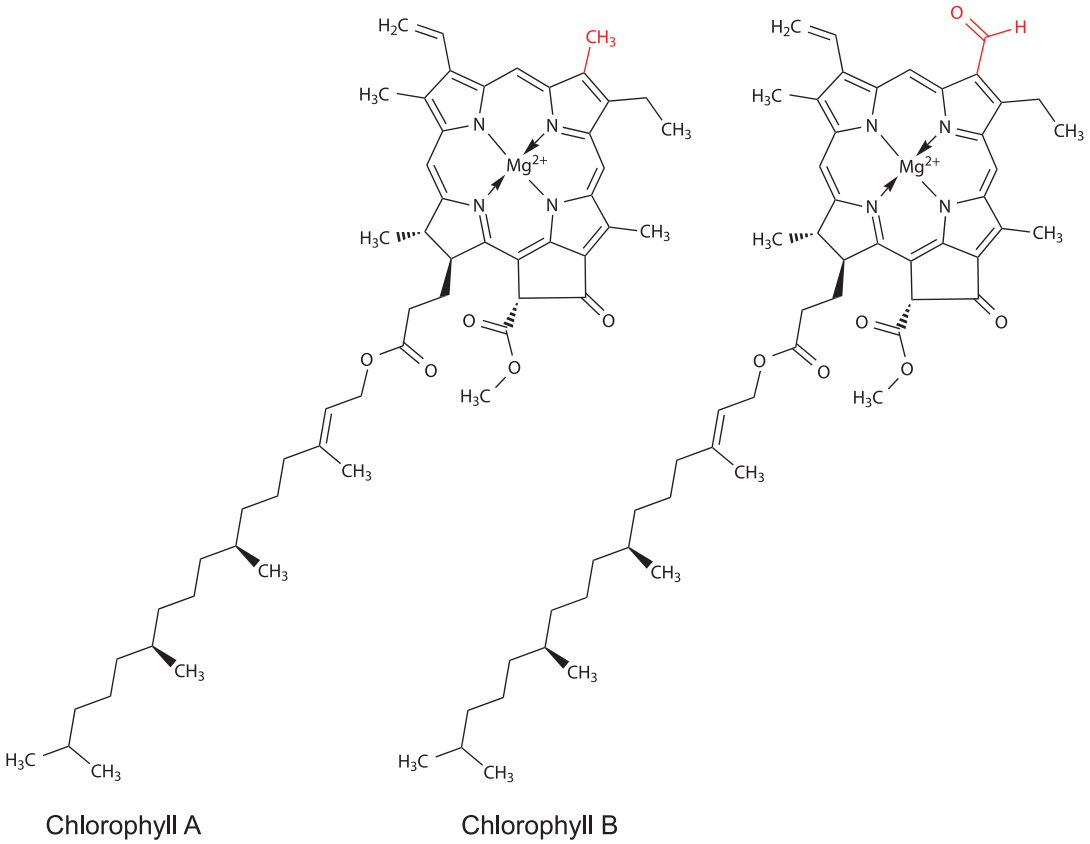


Figure 13.2: Structures of Chlorophyll 'a' and 'b'